WORKED SOLUTION

STEP III 1998, Question 11

Mechanics • Large-Amplitude Pendulum

STEP III 1998, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q11

This question has a full guided treatment in *Beyond STEP*, Section 11.3, *The Real Pendulum*: https://beyondstep.keithmcnulty.org/ch11/real-pendulum.html

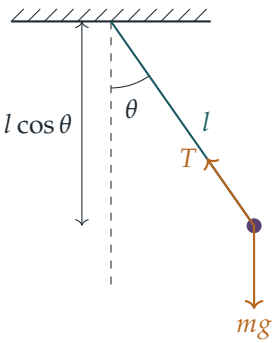
The key idea

Energy conservation gives the angular speed at every angle, and the period is four times the time for a quarter swing, $\int \mathrm{d}t = \int \mathrm{d}\theta/\dot{\theta}$. The half-angle identity $\cos\theta = 1 - 2\sin^2(\theta/2)$ turns the energy equation into the form of the integrand. The suggested substitution removes the awkward infinity at $\theta = \gamma$ and leaves an integrand of the form $(1-x)^{-1/2}$ with x small, which is exactly what the binomial expansion is for.

Solution

The energy equation

Model the pendulum as a particle of mass m on a light inextensible string of length l, swinging in a vertical plane with no air resistance. Measure θ from the downward vertical.



The bob is a depth $l\cos\theta$ below the pivot. The tension acts along the string, perpendicular to the bob's velocity (which is along the arc), so it does no work; only gravity does work, and mechanical energy is conserved. The bob's speed is $l\dot\theta$.

At the extreme angle $\theta=\gamma$ the bob is momentarily at rest. Between there and a general angle θ it falls a height $l\cos\theta - l\cos\gamma$, so

$$\tfrac12 ml^2\dot\theta^2 = mgl(\cos\theta-\cos\gamma) \quad\Longrightarrow\quad \tfrac12 l\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = g(\cos\theta-\cos\gamma),$$

which is the first result.

The period

Using $\cos\alpha = 1-2\sin^2(\alpha/2)$ on both cosines,

$$\cos\theta-\cos\gamma = 2\big(\sin^2(\gamma/2)-\sin^2(\theta/2)\big),$$

so the energy equation becomes

$$\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = \frac{4g}{l}\big(\sin^2(\gamma/2)-\sin^2(\theta/2)\big).$$

By symmetry (the motion is reversible, and the energy equation is unchanged under $\theta\mapsto-\theta$), the four quarter swings $0\to\gamma\to0\to-\gamma\to0$ take equal times. During the quarter swing from $\theta=0$ to $\theta=\gamma$, θ is increasing, so we take the positive square root:

$$\frac{\mathrm{d}t}{\mathrm{d}\theta} = \frac12\sqrt{\frac{l}{g}}\,\big(\sin^2(\gamma/2)-\sin^2(\theta/2)\big)^{-1/2}.$$

Integrating from 0 to γ and multiplying by 4,

$$P = 2\sqrt{\frac{l}{g}}\int_0^\gamma \big(\sin^2(\gamma/2)-\sin^2(\theta/2)\big)^{-1/2}\,\mathrm{d}\theta,$$

as required. (The integrand blows up at $\theta=\gamma$, where the bob is at rest, but the integral converges; the substitution below shows this explicitly.)

The small-amplitude approximation

Write $k=\sin(\gamma/2)$, with $0<\gamma<\pi$ so that $0<k<1$, and put $\sin(\theta/2)=k\sin\phi$. As θ runs from 0 to γ, $\sin\phi$ runs from 0 to 1, so ϕ runs from 0 to $\pi/2$. Differentiating,

$$\tfrac12\cos(\theta/2)\,\mathrm{d}\theta = k\cos\phi\,\mathrm{d}\phi.$$

Since $0\le\theta/2<\pi/2$ and $0\le\phi\le\pi/2$, both cosines are non-negative, so

$$\cos(\theta/2)=\sqrt{1-k^2\sin^2\phi}, \qquad \big(k^2-k^2\sin^2\phi\big)^{1/2} = k\cos\phi.$$

Ask yourself What happens to the troublesome factor $(\sin^2(\gamma/2) - \sin^2(\theta/2))^{-1/2}$ under this substitution?

It becomes $1/(k\cos\phi)$, and the $k\cos\phi$ in $\mathrm{d}\theta$ cancels it exactly:

$$\frac{\mathrm{d}\theta}{\left(k^2 - \sin^2(\theta/2)\right)^{1/2}} = \frac{1}{k\cos\phi}\cdot\frac{2k\cos\phi\,\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}} = \frac{2\,\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}}.$$

Hence

$$P = 4\sqrt{\frac{l}{g}}\int_0^{\pi/2}\left(1-k^2\sin^2\phi\right)^{-1/2}\mathrm{d}\phi,$$

an integral with a perfectly well-behaved integrand. (This much is exact.)

For small γ, $x = k^2\sin^2\phi$ satisfies $0 \le x \le k^2$, which is small, so the binomial expansion $(1-x)^{-1/2} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \cdots$ gives

$$\left(1-k^2\sin^2\phi\right)^{-1/2} \approx 1 + \tfrac{1}{2}k^2\sin^2\phi,$$

with an error of order k^4. Since $\int_0^{\pi/2}\sin^2\phi\,\mathrm{d}\phi = \int_0^{\pi/2}\frac{1}{2}(1-\cos 2\phi)\,\mathrm{d}\phi = \frac{\pi}{4}$,

$$P \approx 4\sqrt{\frac{l}{g}}\left(\frac{\pi}{2} + \frac{k^2}{2}\cdot\frac{\pi}{4}\right) = 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{k^2}{4}\right).$$

Finally, for small γ, $k = \sin(\gamma/2) \approx \gamma/2$ (the error is of order γ^3), so $k^2 \approx \gamma^2/4$ up to terms of order γ^4, and

$$P \approx 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\gamma^2}{16}\right),$$

as required. The leading term is the familiar small-oscillation period; the correction says that larger swings take slightly longer.

Checks

- As $\gamma \to 0$ we recover $2\pi\sqrt{l/g}$, the period from the linearised equation $\ddot{\theta} = -(g/l)\theta$.
- Dimensions: $\sqrt{l/g}$ is a time, and γ is dimensionless.
- Numerically, with $l/g = 1.7/9.81$ and $\gamma = 0.5$, stepping the equation $\ddot{\theta} = -(g/l)\sin\theta$ with a fine Runge–Kutta scheme gives $P = 2.65705$, matching the exact ϕ-integral to all digits shown; the approximation gives 2.65646, a relative error of about 2×10^{-4}. That error scales like γ^4 (it is about $0.0036\gamma^4$), confirming that the γ^2 term is right.

Common slips

- Forgetting the factor 4: the integral from 0 to γ is only a quarter of a period.

- Using $\cos(\theta/2) \approx 1$ in the substitution before expanding. That discards a $k^2 \sin^2\phi$ term of the same order as the correction you are trying to find; keep $\sqrt{1 - k^2\sin^2\phi}$ exact until the binomial step.
- Stopping at $1 + k^2/4$ without converting $k = \sin(\gamma/2)$ into $\gamma/2$.

Beyond STEP

The exact integral $\int_0^{\pi/2}(1 - k^2\sin^2\phi)^{-1/2}\,\mathrm{d}\phi$ is the *complete elliptic integral of the first kind*, $K(k)$, so the exact period is $4\sqrt{l/g}\,K(\sin(\gamma/2))$. Continuing the binomial series gives $1 + \frac{1}{16}\gamma^2 + \frac{11}{3072}\gamma^4 + \cdots$, and Gauss found that K can be computed astonishingly fast with the arithmetic–geometric mean. The fact that the period depends on amplitude is the simplest example of nonlinear oscillation. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats the pendulum and oscillations at first-year level.

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